

CMB8LF v1

COMPLETE PROJECT DOCUMENTATION

Chrystal Moon Base 8-Legged Freaks

Scale	1:3.5 (full-size target 900x700x450 mm)
Legs	8 x 3-servo = 24 servos + 2 gripper servos
Concept	Dual-Master self-repair robot
Material	Prusament PC Space Grade Black
Certification	ESA TML 0.25% - CVCm 0.00%
Target	TRL 5-6 - pitchable to ESA / DLR / private space
Brain	Raspberry Pi 5 8GB + 2x Pico W
Sensors	RPLIDAR A1M8 - BNO055 - VL53L1X x2 - Pi Camera 3
Monitoring	INA226 x4 - TMP117 x2 - DS18B20 x26 (per servo)
License	CC0 2026 - Chrystal Moon Base

PHASE ROADMAP

Phase	Action	Goal
1 - Earth Demo	Build Master 1+2 in PC Space Grade 50+ swap cycles in air - film it	Proof of concept + funding pitch
2 - Vacuum Test	Vacuum chamber at 10 ⁻⁵ mbar Same hardware - no rebuild needed	Validate at ~5000 EUR chamber cost
3 - Field Prep	TRL 5-6 certification Pitch ESA / DLR / private space	First print = flight print

1. VACUUM MATERIAL SYSTEM

1.1 Structural Material

Prusament PC Space Grade Black

Property	Value
ESA Outgassing	TML 0.25% - CVCM 0.00%
Certification	CubeSat + vacuum chamber
Nozzle temp	400 degrees C (hardened steel REQUIRED)
Bed temp	110 degrees C (PEI or glue stick)
Enclosure	Mandatory - heated ~60 degrees C ambient
Fan	OFF during print
Shrinkage	~0.3% (tol_snap = 0.15 mm corrected)
Infill structural	40-60% gyroid
Infill armor	20-30% gyroid
Layer structural	0.2 mm
Layer detail	0.15 mm
Supports	Organic - PC breaks clean

1.2 Vacuum-Rated Consumables

Item	Product	Purpose
Structural adhesive	Master Bond EP21TDCHT	Vac-cert. -196 to +177 deg C
Bearing lube	MoS2 dry (Molykote 321R)	Vacuum-safe. NO grease, NO WD40
Thermal interface	Bergquist GAP PAD 1.0 mm	Servo to armor heat path
Thermal paste	MasterSil 323AO-LO	Low outgassing paste
Radiative coating	Aeroglaze Z306 matte black	Emissivity e=0.95 on armor
Flex parts	PEBA Shore 85 (ASTM E595)	Cable clips, gripper jaws

1.3 Vacuum Qualification Checklist

- ☐ All PC Space Grade prints: 48h bake-out at 85 degrees C minimum
- ☐ Outgassing test per batch (QCM sensor)
- ☐ MoS2 lube: max 1 layer - no overflow
- ☐ EP21TDCHT adhesive: 24h cure at 65 degrees C before vacuum
- ☐ GAP PADs: cut to size - no overlap with connectors
- ☐ Z306 coating: 2 layers - 24h cure - before assembly
- ☐ All cables: NASA-STD-8739.4 harness spec
- ☐ Connectors: gold-plated - crimped - no bare solder
- ☐ QR cycle test: 50x in air - measure pull force after
- ☐ Thermal cycling: -40 to +85 degrees C - 10 cycles - check fit

2. ELECTRONICS & WIRING

2.1 I2C Address Map

Bus	Device	Address	Controller
Pi5 I2C1	BNO055 IMU	0x28	Pi5
Pi5 I2C1	VL53L1X #1 (ToF)	0x29	Pi5
Pi5 I2C1	VL53L1X #2 (ToF)	0x2A	Pi5
Pico #1 I2C0	PCA9685-A (servos)	0x40	Pico W #1
Pico #1 I2C0	PCA9685-B (servos)	0x41	Pico W #1
Pico #2 I2C1	INA226 7.4V Side-A	0x44	Pico W #2
Pico #2 I2C1	INA226 7.4V Side-B	0x45	Pico W #2
Pico #2 I2C1	INA226 6.0V rail	0x46	Pico W #2
Pico #2 I2C1	INA226 5.0V logic	0x47	Pico W #2
Pico #2 I2C1	TMP117 frame temp	0x48	Pico W #2
Pico #2 I2C1	TMP117 battery temp	0x49	Pico W #2
Pi5 1-Wire	DS18B20 x26 (per servo)	—	Pi5 GPIO

2.2 Power Rails

Rail	Source	Consumers	Fuse
11.1V raw	2x LiPo 3S	Buck converter inputs	ATC 50A per pack
7.4V / 20A x4	XL4016 bucks	DS3240 HIP + DS3225 UPPER servos	PTC 30A per buck
6.0V / 10A x2	XL4016 bucks	DS3218 LOWER servos ONLY	PTC 15A per buck
5.0V / 10A x1	Buck converter	Pi5 + sensors + camera + LEDs	PTC 15A

2.3 Servo Channel Map

Leg	Joint	Servo	Voltage	PCA9685	Channel
1 (Front-L ARM)	HIP	DS3240 35kg	7.4V	A (0x40)	0
1 (Front-L ARM)	UPPER	DS3225 25kg	7.4V	A	1
1 (Front-L ARM)	LOWER	DS3218 20kg	6.0V	A	2
1 (Front-L ARM)	GRIPPER	DS3225/MG996R	7.4V	A	12
2 (Front-R ARM)	HIP	DS3240	7.4V	A	3
2 (Front-R ARM)	UPPER	DS3225	7.4V	A	4
2 (Front-R ARM)	LOWER	DS3218	6.0V	A	5
2 (Front-R ARM)	GRIPPER	DS3225/MG996R	7.4V	A	13
3 (Rear-L)	HIP/UPPER/LOWER	DS3240/25/18	7.4/7.4/6V	A	6-7-8
4 (Rear-R)	HIP/UPPER/LOWER	DS3240/25/18	7.4/7.4/6V	A	9-10-11
5-8	HIP/UPPER/LOWER	DS3240/25/18	7.4/7.4/6V	B (0x41)	0-11

2.4 Laser Indicators (Arm Legs 1+2)

Color	Wavelength	Power	GPIO	Meaning
RED	650 nm	5 mW	Pico #1 GP17 (L) / GP19 (R)	Active - sintering mode
GREEN	520 nm	5 mW	Pico #1 GP18 (L) / GP20 (R)	Temperature OK

Laser modules: 6mm diameter - snap fit into tibia and gripper body - no adhesive required

3. PRE-POWER-UP CHECKLIST

Priority	Check
CRITICAL	Measure 7.4V buck WITHOUT load -> must read 7.40V +/- 0.05V
CRITICAL	Measure 6.0V buck WITHOUT load -> must read 5.90V cold / max 6.5V hot
CRITICAL	Measure 5.0V buck -> 5.05V +/- 0.05V
CRITICAL	E-Stop test: relay must cut BOTH packs simultaneously
CRITICAL	PCA9685 V+ pin VISUAL CHECK: must be unconnected on BOTH boards
CRITICAL	DS18B20 sensors: run 1-wire scan, expect 26 devices
NORMAL	i2cdetect -y 1 on Pi5: expect 0x28 0x29 0x2A
NORMAL	Pico #1 I2C0 scan: expect 0x40 0x41
NORMAL	Pico #2 I2C1 scan: expect 0x44 0x45 0x46 0x47 0x48 0x49
NORMAL	Boot Pi5 with zero servo power - verify GPIO 3.3V
NORMAL	Connect servos one leg at a time - watch INA226 current
NORMAL	6.0V HOT test under 2A load: if >6.5V STOP immediately
NORMAL	ARM MODE: QR pull-force test >=20N before first swap
NORMAL	ARM MODE: verify tripod stance lock before arm deploy
NORMAL	Monitor TMP117 + DS18B20 first 10 min of operation

4. BILL OF MATERIALS

4.1 Print Materials

Qty	Component	Spec	Supplier
2 kg	Prusament PC Space Grade Black	ESA TML 0.25% CVCM 0.00%	Prusa Research
—	Hardened steel nozzle 0.4mm	400 deg C rating	E3D / Slice Eng.
as needed	PEBA Shore 85 filament	ASTM E595 vac-safe	Polymaker / Amphora

4.2 Servos (per robot - 26 total)

Qty	Model	Joint	Voltage	Est. Price
8x	DS3240 35kg-cm	HIP	7.4V	~18 EUR/unit
8x	DS3225 25kg-cm	UPPER + FEMUR	7.4V	~14 EUR/unit
8x	DS3218 20kg-cm	LOWER + TIBIA	6.0V	~12 EUR/unit
2x	DS3225 / MG996R	GRIPPER (arms 1+2)	7.4V	~14 EUR/unit

4.3 Electronics

Qty	Component	Address / Pin	Purpose
1x	Raspberry Pi 5 8GB	—	Main SBC
2x	Raspberry Pi Pico W	USB-CDC	Servo ctrl + monitoring
2x	PCA9685 16-ch	0x40 / 0x41	PWM generation
4x	INA226 current sensor	0x44-0x47	Rail monitoring
2x	TMP117 temp sensor	0x48 / 0x49	Frame + battery temp
26x	DS18B20 temp sensor	1-Wire per servo	Thermal cutoff 70 deg C
1x	BNO055 IMU	0x28	9-DOF navigation
2x	VL53L1X ToF	0x29 / 0x2A	Range finding
1x	RPLIDAR A1M8	UART0	360 deg lidar
1x	Pi Camera Module 3	CSI-2 CAM0	12MP HDR camera

4.4 Power System

Qty	Component	Spec	Purpose
2x	LiPo 3S 5000mAh	11.1V nominal	Main energy
2x	BMS 40A 3S	Over-current + balance	Battery protection
4x	Buck 7.4V/20A	XL4016	HIP + UPPER servos
2x	Buck 6.0V/10A	XL4016	LOWER servos
1x	Buck 5.0V/10A	—	Logic + sensors
2x	Anti-spark XT60	—	Battery connectors
4x	PTC fuse 30A	—	7.4V buck outputs
3x	PTC fuse 15A	—	6V + 5V outputs
2x	ATC fuse 50A	—	Battery main leads
1x	30A relay NC	—	E-Stop circuit
1x	NC mushroom button	Red	E-Stop button

4.5 Mechanical

Qty	Component	Spec	Purpose
48x	F688-2RS bearing	8x16x5 mm	All joints
14x	Bearing retention plate	PC Space Grade print	Axial retention

2x	QR connector body	3-pin, >=15A/pin	Leg-swap male
2x	QR connector socket	1.0mm chamfer	Leg-swap female
26x	JST-SM 3-pin pair	—	Servo connectors
~120x	M3 heat-set inserts	6.0 mm deep	Screw anchors
assorted	M3 bolts	various lengths	Assembly

4.6 Vacuum-Rated Materials

Product	Purpose	Est. Price
Master Bond EP21TDCHT (100g kit)	Structural adhesive vac-cert.	~80 EUR
Molykote 321R MoS2 spray	Bearing lube - no grease in vacuum	~25 EUR
Bergquist GAP PAD 1.0mm (A4 sheet)	Thermal interface servo to armor	~30 EUR
MasterSil 323AO-LO	Low-outgassing thermal paste	~45 EUR
Aeroglaze Z306 matte black (spray)	Radiative cooling e=0.95 on armor	~35 EUR

4.7 Lighting

Qty	Component	GPIO	Purpose
1x	WS2812B ring 24 LEDs	Pico #1 GP14	Body accent
2x	WS2812B RGB eye	chain GP14	Head eyes
1x	10W Spotlight	GP21 MOSFET	Head front
2x	3W White LED	GP20 MOSFET	Head sides
4x	Laser red 650nm 5mW 6mm	GP17 / GP19	Arm targeting
4x	Laser green 520nm 5mW 6mm	GP18 / GP20	Arm temp OK

4.8 Cost Estimate

Category	Per Robot	Dual-Master (x2)
Print material (PC Space Grade)	~100 EUR	~200 EUR
Servos (26 units)	~330 EUR	~660 EUR
Electronics	~320 EUR	~640 EUR
Power system	~150 EUR	~300 EUR
Mechanical	~80 EUR	~160 EUR
Vacuum materials	~215 EUR	~430 EUR
Lighting	~50 EUR	~100 EUR
TOTAL	~1,245 EUR	~2,490 EUR

Vacuum chamber test Phase 2: ~5,000 EUR chamber time

5. SHOPPING LIST

5.1 Print Materials

Product	Link / Notes	Price
Prusament PC Space Grade Black 1kg	prusa3d.com/product/prusament-pc-space-grade-black-1kg	~45 EUR/kg
Hardened steel nozzle 0.4mm E3D V6	e3d-online.com/products/nozzle-x-v6	~15 EUR
PEBA Shore 85 filament	Amphora AM3300 or Polymaker PolyFlex (vac-safe)	~35 EUR/kg

5.2 Servos

Product	Link / Notes	Price
DS3240 35kg x8	aliexpress.com/item/1005003822936048	~18 EUR/unit
DS3225 25kg x10 incl. grippers	aliexpress.com/item/4000083855274	~14 EUR/unit
DS3218 20kg x8	aliexpress.com/item/32996186179	~12 EUR/unit

5.3 Key Electronics

Product	Link / Notes	Price
Raspberry Pi 5 8GB	raspberrypi.com/products/raspberry-pi-5	~85 EUR
Raspberry Pi Pico W x2	raspberrypi.com/products/raspberry-pi-pico	~8 EUR/unit
INA226 breakout x4	aliexpress.com/item/32785084311	~3 EUR/unit
DS18B20 waterproof x26	aliexpress.com/item/32523899954	~1.50 EUR/unit
TMP117 breakout x2	adafruit.com/product/4821	~12 EUR/unit
RPLIDAR A1M8	slamtec.com/en/Lidar/A1	~90 EUR
Pi Camera Module 3	raspberrypi.com/products/camera-module-3	~30 EUR

5.4 Vacuum Materials

Product	Link / Notes	Price
Master Bond EP21TDCHT 100g kit	masterbond.com/tds/ep21tdcht	~80 EUR
Molykote 321R MoS2 spray	rs-components.com - search Molykote 321R	~25 EUR
Bergquist GAP PAD 1.0mm A4	bergquistcompany.com	~30 EUR/sheet
MasterSil 323AO-LO	masterbond.com/tds/mastersil-323ao-lo	~45 EUR
Aeroglaze Z306 spray	henkel-adhesives.com - search Aeroglaze Z306	~35 EUR

5.5 Mechanical

Product	Link / Notes	Price
F688-2RS bearings x48+	farnell.com - search F688-2RS	~2 EUR/unit
M3 heat-set inserts x120	aliexpress.com/item/4000232858343	~10 EUR (100pcs)
JST-SM 3-pin pairs x26	aliexpress.com/item/32674084915	~0.50 EUR/pair

5.6 Lighting & Lasers

Product	Link / Notes	Price
Laser red 650nm 6mm x4	amazon.nl/dp/B0982CYKYV	~8 EUR (10pcs)
Laser green 520nm 6mm x4	ebay.com/itm/203590420513	~4 EUR/unit
WS2812B ring 24 LED	aliexpress.com/item/32958819003	~5 EUR
10W spotlight module	aliexpress.com/item/32956099804	~6 EUR

Order Priority

1. Direct: Prusament PC Space Grade - DS18B20 x26 - F688-2RS bearings
2. Then: Servos - electronics - power components

3. When prints ready: EP21TDCHT - MoS2 - GAP PADS - Z306
4. Last: Lighting (after fit validation of chassis)

6. OPENS CAD FILE SUMMARY

All files use Prusament PC Space Grade Black. Nozzle: 400 deg C hardened steel. Enclosure mandatory. MoS2 dry lube on all bearings. No grease.

CMB8LF_v1_vacuum_config.scad

Central config file. Include in all modules.

```
MATERIAL = "PC-SpaceGrade"
VACUUM_RATED = true
PHASE = 1 // 1=Earth demo, 2=Vacuum, 3=Field
NOZZLE_TEMP = 400 // deg C
BED_TEMP = 110 // deg C
SERVO_TEMP_CUTOFF = 70 // deg C per DS18B20
QR_ALIGNMENT_TOL = 2.0 // mm for autonomous docking
```

CMB8LF_v1_leg_complete.scad

Standard leg: hip + femur + tibia. 3 segments, 3 servos.

```
SHOW_SKELETON = true/false // PC Space Grade structural
SHOW_ARMOR = true/false // PC Space Grade armor
SHOW_THERMAL = true/false // Debug: cooling channels
SHOW_ASSEMBLY = true/false // Full assembled view
Modules: hip_skeleton() femur_skeleton() tibia_skeleton()
Armor: hip_armor() femur_armor() tibia_armor()
Bearings: F688-2RS 8x16x5mm - MoS2 dry lube only
```

CMB8LF_v1_arm_leg_combined.scad

Legs 1+2 (Front arms). Dual-function: walk OR manipulate.

```
MODE = "assembly" / "arm" / "leg" / "test"
femur_len = 134mm // +12mm vs standard for arm reach
Includes: gripper_body() gripper_jaw() gripper_cam()
Includes: gripper_pushrod() qr_connector_body()
Includes: qr_connector_socket() with 1.0mm chamfer
Laser mounts: 6.2mm hole in tibia + gripper body
QR: 3-pole, >=15A/pin, >=20N pull force, +-2mm align tol
```

CMB8LF_v1_head_module.scad

Head with all sensors, lights and camera.

```
MODE = "assembly" / "skeleton" / "lid" / "test"
RPLIDAR dome: 68mm diameter, 4x M3 mounts
Pi Camera: 25mm module, +/-30 deg tilt, CSI ribbon slot
VL53L1X x2: front + downward, XSHUT accessible
WS2812B ring: 88mm OD, snaps into groove
RGB eyes x2: 12mm bezel, WS2812B chain
Spotlight 10W + 3W LED x2: MOSFET-switched
Laser x4: 6.2mm holes front corners (R+G each side)
Neck: 42mm tube, 4x M3, +-15 deg tilt (nod)
```

CMB8LF_v1_body_chassis.scad

Main body. Octagonal. All electronics + power inside.

```
MODE = "assembly"/"skeleton"/"lid_top"/"lid_bottom"/"armor"/"test"
body_w=240mm body_d=200mm body_h=55mm (octagonal)
Component pockets: Pi5 - Pico x2 - PCA9685 x2
Power pockets: 7x buck - 2x BMS - 2x LiPo - relay
Star-point GND block: central - all GNDs here only
8x leg mount interfaces: QR socket at each leg point
Radial cable channels to all 8 leg mounts
Neck mount top: 42mm pipe to head module
Armor: 2x side panels, snap-fit dovetail
```

7. ASSEMBLY ORDER

7.1 Body (chassis)

1. Star-point GND block: mount centrally. All GND cables connect here ONLY.
2. BMS x2: left and right. Connect before inserting batteries.
3. LiPo x2: slide in through bottom lid battery access hatch.
4. Buck converters x7: set voltages BEFORE connecting servos.
5. 7.4V bucks: set to exactly 7.40V (no load)
6. 6.0V bucks: set to 5.90V cold. Max 6.5V hot under 2A load.
7. 5.0V buck: set to 5.05V
8. E-Stop relay: front central. Test NC circuit before proceeding.
9. PCA9685 x2: V+ pin MUST be unconnected. Verify visually.
10. Pico W x2: connect via USB to Pi5.
11. Pi5: last. USB-C from 5V buck. Verify 5.05V before connecting.
12. 8x leg mount QR sockets: route cables through channels first.
13. Side armor panels: dovetail snap-fit.
14. Lids top + bottom: 8x M3 bolts each.

7.2 Each Leg

1. Apply MoS2 dry lube to F688-2RS bearings before press-fit.
2. Press bearings into pockets (H7 fit, 0.1mm tolerance).
3. Install M3 heat-set inserts: 200 deg C soldering iron, 6mm depth.
4. Mount servos: DS3240 hip, DS3225 upper, DS3218 lower.
5. Apply Bergquist GAP PAD 1.0mm on servo seat (thermal path to armor).
6. Route cables through cable channels. Install PEBA cable clips.
7. Snap armor panels onto dovetail rails.
8. Install bearing retention plates (M3 bolt each).
9. Slide in laser modules (red 650nm + green 520nm). No adhesive.
10. Connect leg QR body to chassis QR socket. Test 20N pull force.

7.3 Head

1. Mount RPLIDAR A1M8 dome: 4x M3 heat-set, cable via dedicated slot.
2. Install BNO055 IMU: central top, tape + 2x M2.
3. Install VL53L1X x2: XSHUT cables accessible for address config.
4. Pi Camera Module 3: front central, CSI ribbon through neck channel.
5. WS2812B ring: press into groove (88mm OD). Data from Pico GP14.
6. RGB eyes x2: press into 12mm bezels. Daisy-chain WS2812B.
7. Spotlight 10W + 3W LED x2: MOSFET-switched, heatsink on 10W.
8. Laser indicators x4: slide into 6.2mm holes front corners.
9. Front armor panel: snap onto dovetail.
10. Lid: 4x M3 bolts.
11. Connect neck to body: 4x M3 bolts. Route all cables through tube.

8. DUAL-MASTER SELF-REPAIR CONCEPT

Both Master 1 and Master 2 are identical full-spec robots. Neither is a passive dummy. Both are built to vacuum-rated PC Space Grade spec.

8.1 Leg-Swap Procedure

- Master 2 identifies damaged leg via INA226 (overcurrent) or DS18B20 (overtemp).
- Master 2 enters tripod stance on Legs 3-8. Verify INA226 <5A per leg.
- Master 1 approaches with arm-leg (Leg 1 or 2) extended.
- Master 1 gripper aligns with QR socket on Master 2 damaged leg.
- QR chamfer (1.0mm) allows +/-2mm alignment error for autonomous docking.
- Master 1 pulls failed leg free (>20N force threshold, 3-pole lock releases).
- Master 1 positions spare leg and inserts into Master 2 QR socket.
- Lock pins engage. Master 2 verifies servo comms on new leg.
- Both robots resume normal operation.

8.2 Safety Protocols

Protocol	Description
Thermal cutoff	DS18B20 per servo reports to Pi5. Auto-shutdown at 70 deg C.
Current limit	INA226 on every rail. Alert thresholds per servo group.
E-Stop	Physical NC relay cuts BOTH packs. Hardware, not software.
Hibernation	Pi5 shuts non-essentials if power drops below threshold.
Stance lock	Software must confirm tripod stance before arm deploy.
QR pull test	>=20N pull force validated every 50 swap cycles.

CMB8LF v1 - Chrystal Moon Base 8-Legged Freaks - Vacuum-Rated - Dual-Master Self-Repair - Scale 1:3.5 - CC0 2026
First print = Flight print. PC Space Grade certified for CubeSat use from day one.